Design Documentation

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Design Documentation – Interview Vault App

Application Description and Target User

The Interview Vault app is a mobile application built using React Native and Expo. It helps users track their job applications, interviews, and progress during their job search.

Users can add job applications, update their status, view details, and delete them when needed. The app also shows a dashboard with statistics and interview tips.

The target users are:

- Students looking for internships
- Job seekers
- Recent graduates tracking multiple applications

Functional and Non-Functional Requirements

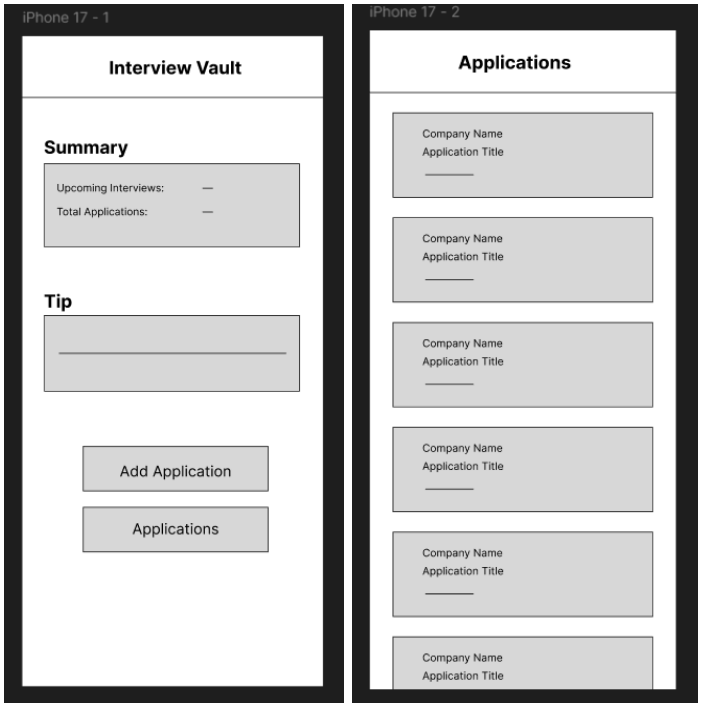
Functional Requirements

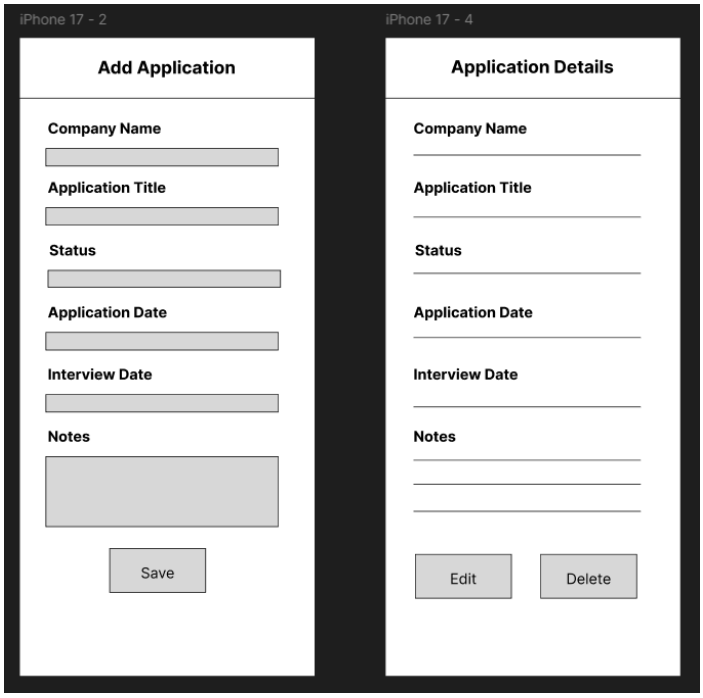
- User can add a new job application
- User can edit an existing application
- User can delete an application
- User can view all applications in a list
- User can view full details of one application
- User can track application status (Applied, Interview, Offer, etc.)
- App shows total applications and upcoming interviews
- App fetches a random interview tip from an API

Non-Functional Requirements

- App must be fast and responsive
- UI should be simple and easy to use
- Data should persist using local storage (AsyncStorage)
- App should handle errors without crashing
- Navigation should be smooth between screens

Wireframes and User Flow





Component Hierarchy and Data Flow

Component Structure

- AppNavigator (handles navigation)
    - HomeScreen
    - ApplicationsScreen
        - ApplicationCard (list item)
        - EmptyState
        - LoadingState
    - AddEditApplicationScreen
    - ApplicationDetailsScreen

Data Flow

- Data is stored locally using AsyncStorage
- When user adds/edits:
  → data is saved using storage functions
- When screens load:
  → data is fetched and displayed
- Navigation passes data between screens:
    - Example: Application object passed to Details screen

Description of Major Design Decisions

The app uses some main design decisions to make it work better and easier to use. First I used AsyncStorage because it is simple and I dont need a backend and its good for small apps like this. Then I used stack navigation so the user can move between screens like Home, list, details and edit easily and it makes the app feel simple. I also added form validation to stop errors, like making sure the company and title are not empty and the date is in the right format (YYYY-MM-DD). I created reusable components like ApplicationCard, LoadingState and EmptyState so the code is more clean and easier to manage later. The app also uses a status system like Applied, Interview Scheduled, Offer Received and Rejected so the user can track their progress better. Finally I added a simple API for interview tips so the app is not boring and gives useful tips to the user.

Contribution Summary

Mike – Project Setup, Navigation, Integration

Mike took the lead on the overall structure of the project. He created the Expo app, installed dependencies, organized the folder structure, built the navigation system, developed the Home screen, connected all screens together, and handled final integration and bug fixing. His role helped ensure the project had a strong foundation and that the full app worked smoothly as one complete build.

Ibrahim – API Integration and Applications Screen

Ibrahim was responsible for choosing and connecting the REST API used on the Home screen. He also created the API service file and handled loading, error, and empty states. In addition, he built the Applications screen and connected stored job application data to the list interface. His work helped bring both live data and clear application tracking into the app.

Shumail – AsyncStorage and Add/Edit Application Screen

Shumail worked on the data storage side of the app. He created the AsyncStorage helper file and built the Add/Edit Application screen, including the form fields and the logic for saving and updating applications. His contribution gave the app its core functionality by allowing users to add, edit, and manage their own information.

Natasha – Details Screen, Design Support, Documentation

Natasha developed the Application Details screen and implemented the delete flow. She also contributed to the wireframes, styling polish, screenshots, testing documentation, bug summaries, and formatting of the final materials. Her work helped improve both the presentation and completeness of the final submission.

Team Collaboration Through GitHub

As a group, we used GitHub to manage our workflow and combine everyone's work efficiently. Each member focused on their assigned part of the app, which allowed development to happen in parallel. Once the individual features were completed, the project was merged into one final version, followed by testing, refinement, and bug fixing. This workflow helped us collaborate smoothly and ensured the final app felt cohesive and fully connected.

Testing Documentation

Mobile Application Testing Documentation

Application Name: Interview Vault

Version: 1.0.0

Platform: Android (React Native / Expo)

Document Date: April 13, 2026

Testing Environment: Android Emulator

1. Application Overview

Interview Vault is a mobile application designed to help job seekers track and manage their job applications. Users can add, edit, view, and delete job applications while tracking status updates.

Key Features

Feature	Description
Dashboard	Displays total applications count, upcoming interviews, and motivational quotes
Application List	View all tracked applications with color-coded status badges
Add/Edit Application	Create or modify application entries with company, title, status, dates, and notes
Application Details	View full details with options to edit or delete
Data Persistence	AsyncStorage for offline data persistence

Technical Stack

Component	Technology
Framework	React Native 0.83.4 / Expo SDK 55
Navigation	React Navigation 7.x (Native Stack)
Storage	AsyncStorage 2.2.0
Language	TypeScript 5.9.2

2. Comprehensive Test Plan

ID	Category	Test Case	Steps to Execute	Expected Result	Status
TC-001	Navigation	App Launch	1. Launch the application	Home screen displays with title, dashboard, and navigation buttons	**PASS**
TC-002	Navigation	Screen Navigation Flow	1. Tap "View Applications" 2. Tap application card 3. Use back button	Navigates correctly between all screens and returns properly	**PASS**

TC-003	Navigation	FAB Navigation	1. On Applications screen 2. Tap floating + button	Navigates to Add Application screen	**PASS**
TC-004	Data - Create	Add New Application	1. Navigate to Add Application 2. Fill all fields 3. Tap "Add Application"	Application saved and appears in list	**PASS**
TC-005	Data - Create	Required Fields Only	1. Fill Company, Title, App Date only 2. Save	Application saved with optional fields empty	**PASS**
TC-006	Data - Read	View Application List	1. Create multiple applications 2. Navigate to Applications screen	All applications displayed sorted by newest first	**PASS**
TC-007	Data - Read	View Application Details	1. Tap on application card	All fields displayed correctly on details screen	**PASS**
TC-008	Data - Update	Edit Application	1. Open details 2. Tap Edit 3. Modify fields 4. Save	Changes saved and reflected in list/details	**PASS**

TC-009	Data - Delete	Delete Application	1. Open details 2. Tap Delete 3. Confirm deletion	Application removed, confirmation dialog shown first	**PASS**
TC-010	Data - Persistence	Data Persists After Restart	1. Create applications 2. Close app 3. Reopen app	All applications persist and display	**PASS**
TC-011	Validation	Empty Required Fields	1. Leave Company or Title empty 2. Tap Save	Validation errors displayed, form not submitted	**PASS**
TC-012	Validation	Invalid Date Format	1. Enter date as "13/04/2026" 2. Tap Save	Error: "Use YYYY-MM-DD format."	**PASS**
TC-013	UI/UX	Status Badge Colors	1. Create apps with different statuses 2. View in list	Each status has distinct color (Applied=blue, Rejected=red, etc.)	**PASS**
TC-014	UI/UX	Empty State Display	1. Delete all applications 2. View Applications screen	Empty state message: "No applications yet"	**PASS**
TC-015	UI/UX	Loading States	1. Navigate to screens 2. Observe during data load	Loading spinner displayed while fetching data	**PASS**

TC-016	API	Quote Loads on Home	1. Navigate to Home screen	Motivational quote fetched and displayed	**PASS**
TC-017	API	API Offline Fallback	1. Disable network 2. Navigate to Home	Fallback quote displayed instead of error	**PASS**

3. Bug Summary & Resolution Log

Bug ID	Severity	Description	Root Cause	Resolution	Status
BUG-001	High	Application Details Screen Errors - Screen threw errors and did not display properly	Component implementation issues in ApplicationDetailsScreen.tsx caused rendering errors	Fixed by npineault (commit 67e1fb5) - Rebuilt screen with proper structure	**RESOLVED**
BUG-002	High	Merge Conflicts in Components - Multiple team members' implementations conflicted	Parallel development on overlapping files without proper branch coordination	Resolved by Shumail (commit 0a9c5f3) - Kept consistent component versions	**RESOLVED**

BUG-003	Medium	Expo Entry Point Setup Issue - App failed to start due to incorrect configuration	Initial project setup missing proper Expo entry point in package.json	Fixed by Michael Morrison (commit c26a761) - Added correct entry configuration	**RESOLVED**

Bug Statistics

Metric	Count
Total Bugs Identified	3
Bugs Resolved	3
Bugs Outstanding	0

4. Screenshots

** Refer to screenshots Tab **

4.1 Home Screen

Screenshot 4.1 - Home Screen

Insert screenshot showing: Dashboard with title, summary card, statistics, quote card, and navigation buttons

4.2 Applications List

Screenshot 4.2 - Applications List

Insert screenshot showing: Application cards with company names, titles, and color-coded status badges

4.3 Add Application Form

Screenshot 4.3 - Add Application

Insert screenshot showing: Form with all fields (Company, Title, Status chips, Dates, Notes)

4.4 Form Validation

Screenshot 4.4 - Validation Errors

Insert screenshot showing: Form with validation error messages displayed

4.5 Application Details

Screenshot 4.5 - Application Details

Insert screenshot showing: Full application details with Edit and Delete buttons

4.6 Delete Confirmation

Screenshot 4.6 - Delete Confirmation Dialog

Insert screenshot showing: Alert dialog with Cancel and Delete options

4.7 Empty State

Screenshot 4.7 - Empty State

Insert screenshot showing: Empty applications list with "No applications yet" message

5. Test Summary

Category	Tests	Passed	Pass Rate
Navigation	3	3	**100%**
Data Handling (CRUD)	7	7	**100%**

Validation	2	2	**100%**
UI/UX	3	3	**100%**
API Integration	2	2	**100%**
TOTAL	**17**	**17**	**100%**

Conclusion

The Interview Vault application has been tested across navigation, data handling (CRUD operations), form validation, UI/UX, and API integration. All 17 test cases passed successfully with a 100% pass rate. The three bugs identified during development have been resolved, and the application is functioning as expected on the Android Emulator.

--- End of Testing Documentation ---

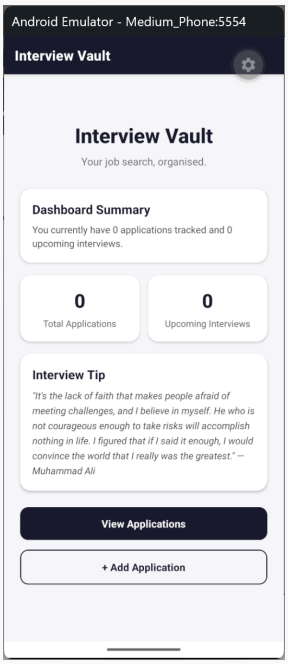
Screenshots

4. Screenshots

4.1 Home Screen

Screenshot 4.1 - Home Screen

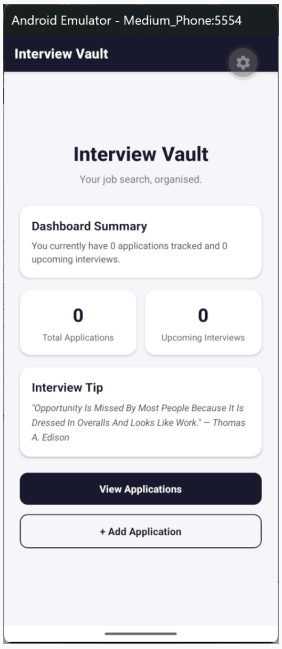
Insert screenshot showing: Dashboard with title, summary card, statistics, quote card, and navigation buttons



4.2 Applications List

Screenshot 4.2 - Applications List

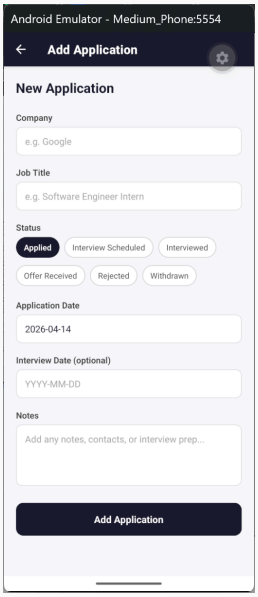
Insert screenshot showing: Application cards with company names, titles, and color-coded status badges



4.3 Add Application Form

Screenshot 4.3 - Add Application

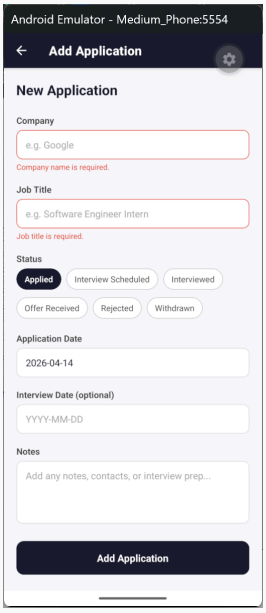
Insert screenshot showing: Form with all fields (Company, Title, Status chips, Dates, Notes)



4.4 Form Validation

Screenshot 4.4 - Validation Errors

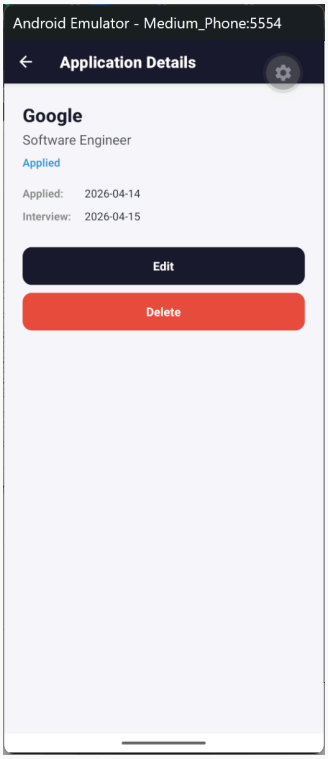
Insert screenshot showing: Form with validation error messages displayed



4.5 Application Details

Screenshot 4.5 - Application Details

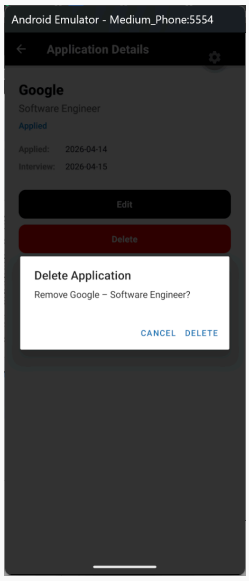
Insert screenshot showing: Full application details with Edit and Delete buttons



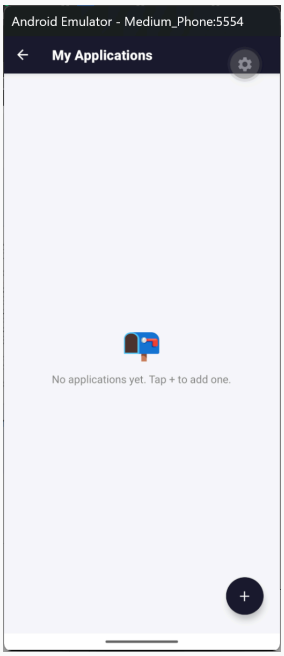
4.6 Delete Confirmation

Screenshot 4.6 - Delete Confirmation Dialog

Insert screenshot showing: Alert dialog with Cancel and Delete options



4.7 Empty State

Screenshot 4.7 - Empty State

Insert screenshot showing: Empty applications list with "No applications yet" message